PCA_uflo

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Setup

```
library(tidyverse)
```

```
## ── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
## ✔ dplyr     1.2.1     ✔ readr     2.2.0
## ✔ forcats   1.0.1     ✔ stringr   1.5.1
## ✔ ggplot2   4.0.3     ✔ tibble    3.3.1
## ✔ lubridate 1.9.5     ✔ tidyr     1.3.2
## ✔ purrr     1.0.2
## ── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()    masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts
to become errors
```

```
setwd("/home/FM/sblanco/GWAS_8072026/uflo/05_filtering")
```

Read in data

Eigenvec and eigenval files generated from a filtered VCF file using Plink. VCF filters included: QUAL=30; MIN_DEPTH=10; MAX_DEPTH=240; MAF=0.10; MISS=0.75. Filtered VCF had 10,962 SNPs (aligned to uflo ref genome). After pruning, 9,953 of 10,962 variants were removed, leaving 1,009 remaining variants.

```
pca <-read_table("uflo_75_pruned.eigenvec", col_names = FALSE)
```

```
##
## ── Column specification ────────────────────────────────────────────────────────
## cols(
##   .default = col_double(),
##   X1 = col_character(),
##   X2 = col_character()
## )
## ℹ Use `spec()` for the full column specifications.
```

```
eigenval <- scan("uflo_75_pruned.eigenval")
```

Clean data

Remove nuisance column and set names

```
pca <- pca[,-1]
names(pca)[1] <- "ind"
names(pca)[2:ncol(pca)] <- paste0("PC", 1:(ncol(pca)-1))
```

Assign labels

Species labels

```
spp <- rep(NA, length(pca$ind))
spp[grep("UFLO", pca$ind)] <- "Usnea florida"
spp[grep("USUB", pca$ind)] <- "Usnea subfloridana"
```

Locations labels

```
loc <- rep(NA, length(pca$ind))
loc[grep("11_0", pca$ind)] <- "Russia"
loc[grep("09_", pca$ind)] <- "Latvia"
loc[grep("UFLO_0", pca$ind)] <- "Latvia"
loc[grep("10_0", pca$ind)] <- "Latvia"
loc[grep("08_", pca$ind)] <- "Latvia"
loc[grep("14", pca$ind)] <- "Belarus"
loc[grep("FR", pca$ind)] <- "Austria"
loc[grep("04_0", pca$ind)] <- "Estonia"
loc[grep("05_", pca$ind)] <- "Estonia"
loc[grep("06_", pca$ind)] <- "Estonia"
loc[grep("FG", pca$ind)] <- "Austria"
```

Remake data.frame

```
pca <- as_tibble(data.frame(pca, spp))
```

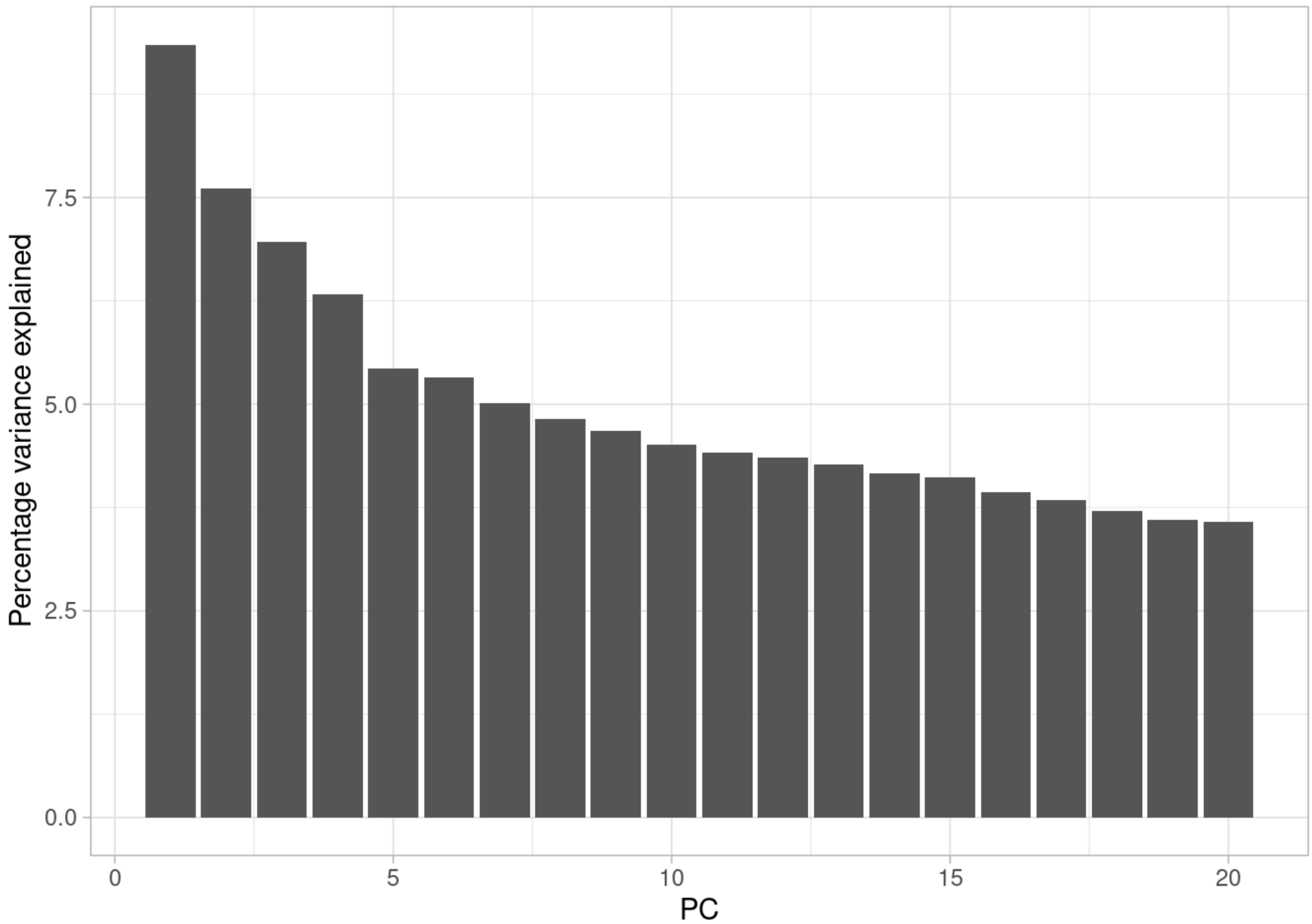
Plot variance

Convert to percentage variance explained

```
pve <- data.frame(PC = 1:20, pve = eigenval/sum(eigenval)*100)
```

Plot pve

```
ggplot(pve, aes(PC, pve)) + geom_bar(stat = "identity") + ylab("Percentage variance expl
ained") + theme_light()
```



Calculate the cumulative sum of the percentage variance explained

```
cumsum(pve$pve)
```

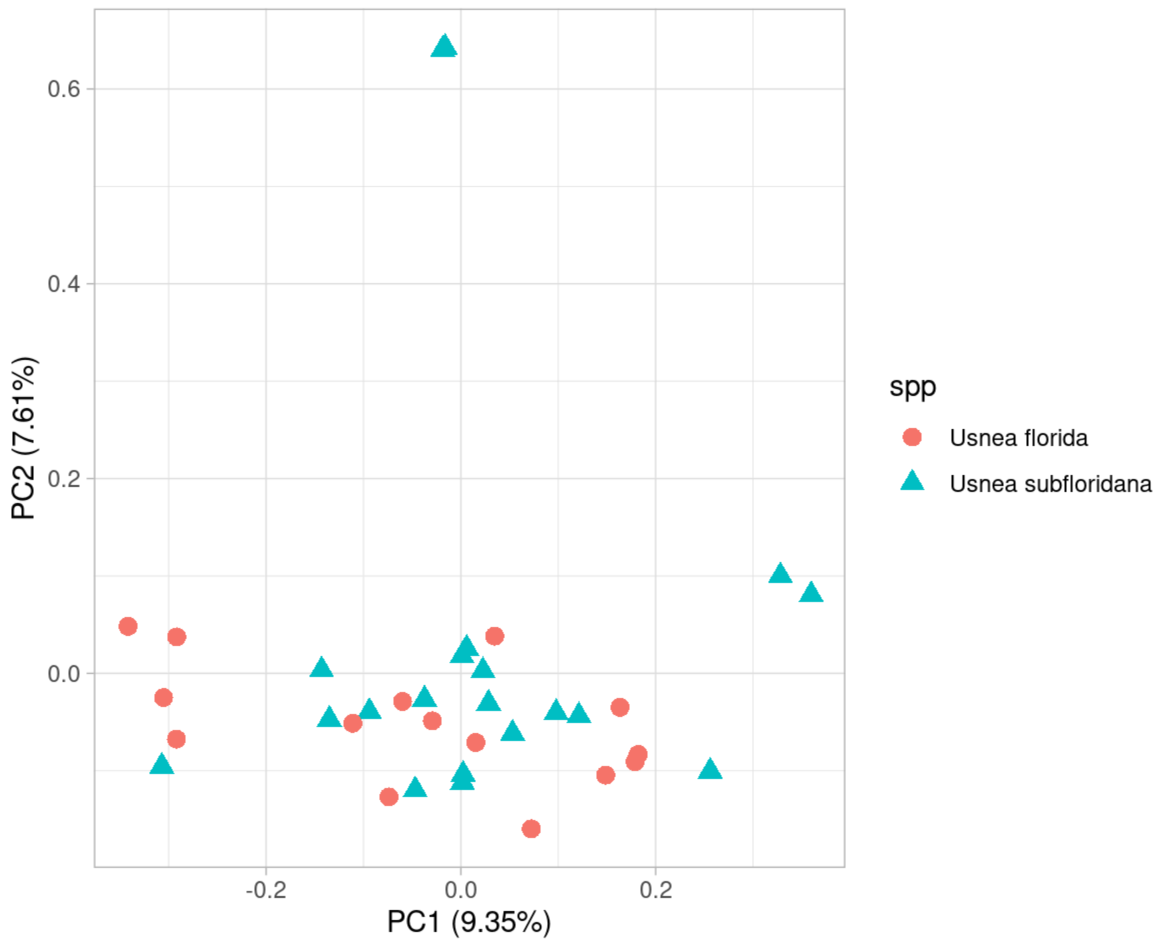
```
##  [1]   9.349512  16.959193  23.918516  30.249329  35.676577  40.997040
##  [7]  46.014629  50.832185  55.506064  60.015008  64.434974  68.789508
## [13]  73.063319  77.226723  81.346905  85.277272  89.117212  92.827374
## [19]  96.428844 100.000000
```

> PC1 & PC2 explain ~17% of the variance observed

Plot PCA

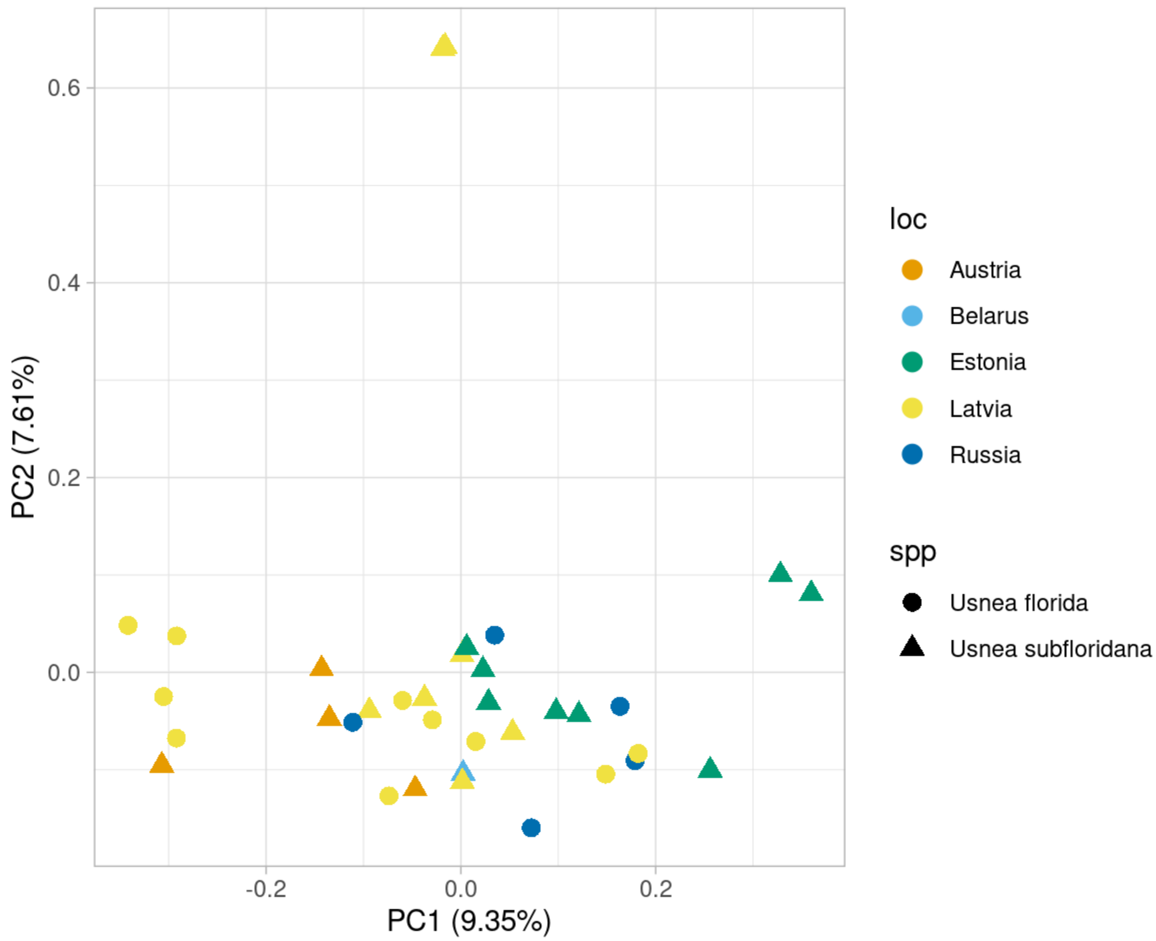
By species

```
ggplot(pca, aes(PC1, PC2, shape = spp, col=spp)) + geom_point(size = 3) + coord_equal()
+ theme_light() + xlab(paste0("PC1 (", signif(pve$pve[1], 3), "%)")) + xlab(paste0("PC1
(", signif(pve$pve[1], 3), "%)")) + ylab(paste0("PC2 (", signif(pve$pve[2], 3), "%)"))
```



By location

```
ggplot(pca, aes(PC1, PC2, shape = spp, col=loc)) + geom_point(size = 3) + scale_colour_m
anual(values = c("#E69F00", "#56B4E9", "#009E73", "#F0E442", "#0072B2")) + coord_equal()
+ theme_light() + xlab(paste0("PC1 (", signif(pve$pve[1], 3), "%)")) + xlab(paste0("PC1
(", signif(pve$pve[1], 3), "%)")) + ylab(paste0("PC2 (", signif(pve$pve[2], 3), "%)"))
```



No PCA structure based on species or location